

A Practical View of Research Methodology



By

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Speaker's Profile



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- ❧ Research Interests:
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 - ❧ Internet of Things (IoT)



PART-1



What is 'Research' and its components?

Part 1 – Understanding



- ❧ What is “Research”?
- ❧ Why do we need research?
- ❧ Who would do a research?

Research Origin



❧ Al-Hasan Bin Al-Haytham Al-Basri (965 - 1040 A.C.)

- ❧ He set the very first systematic principles of the scientific researches those still in use.
- ❧ He was the very first scientist who created lens, those used for magnification and for eyeglasses during his era.



What is RESEARCH?



1. The systematic investigation into and study of materials and sources in order to establish facts and reach new conclusions.
2. Investigate systematically.
3. Meriam-Webster-1: careful study that is done to find and report new knowledge about something
4. Meriam-Webster-2: the activity of getting information about a subject

Similar Words



∞ Research

∞ Replay

∞ React

∞ Rejoin

∞ Retry

.

.

.

Etc.

MATTER vs PROBLEM

Matter is yet to be a Problem

Search
for a
Matching-Key



No
Key

Re-search
for a
Solution

Before the matter becomes a problem, we searched for a proper matching-key, when we did not find that key, then the matter have transformed to be a problem.

- Every Problem, originally was a matter.

What is RESEARCH?

Exploring / Seeking for better solutions.



What is RESEARCH?



A research can be defined with either one of the following:

- ❧ A research is a systematic investigation to find new/different ways of solving a problem.
- ❧ A research is a systematic investigation to seek new knowledge about some topic.

An Example



Scenario Setup

Two guys are thirsty. They have one single apple to share. And because they believe that they are equal to each other in term of rights, they have decided to divide the apple into two parts those must be as identical as they can (in quantity) to have it to quench their thirst.

Problem?

Problem We want to solve

Aim?

What solves the problem?

Objective

What to do to achieve the aim(s).

Scope?

?

Methodology?

?

An Example



Scenario Setup

Two guys are thirsty. They have one single apple to share. And because they believe that they are equal to each other in term of rights, they have decided to divide the apple into two parts those must be as identical as they can (in quantity) to have it to quench their thirst.

Problem?

The two guys are thirsty, got no water but got an apple, and they are not sure how to divide it equally.

Aim?

To quench their thirst equally with an apple.

Objective

1. To determine the center of the apple.
2. To divide the apple into two identical parts (in quantity).

Scope?

?

Methodology?

?

An Example



❧ Problem?

❧ Problem We want to solve

❧ Aim?

❧ What solves the problem?

❧ Objective

❧ What to do to achieve the aim(s).

❧ Scope?

❧ ?

❧ Methodology?

❧ ?

❧ Problem?

❧ The two guys are thirsty, got no water but got an apple, and they are not sure how to divide it equally.

❧ Aim?

❧ To quench their thirst equally with an apple.

❧ Objective

- ❧ 1. To determine the center of the apple.
- ❧ 2. To Divide the apple into two identical parts (in quantity).

❧ Scope?

❧ ?

❧ Methodology?

❧ ?



Research Methodology

Method Vs. Methodology



Vertical Knife-Cutting



Vertical Laser-Cutting

Horizontal Knife-Cutting



Horizontal Laser-Cutting

Blending
The
Apple

Research Methodology



Time

Trigger
Event
&
Input

Method-1

O/p1

Method-2

O/p2

Method-3

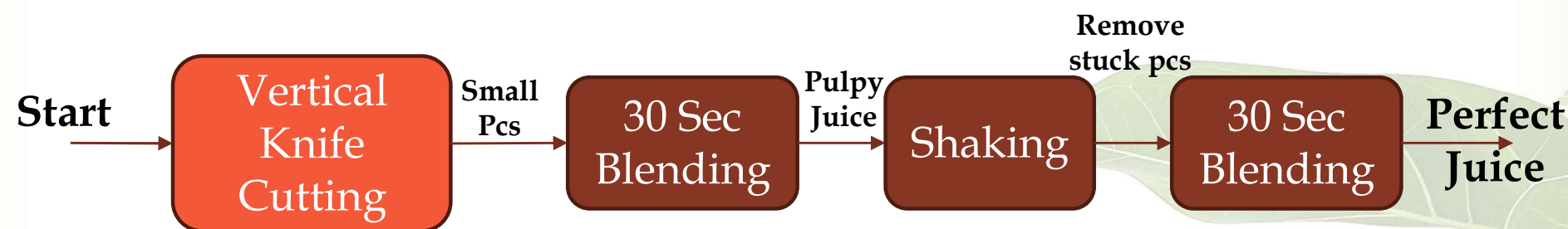
Final
Output

A Methodology

R.M. Example



Time



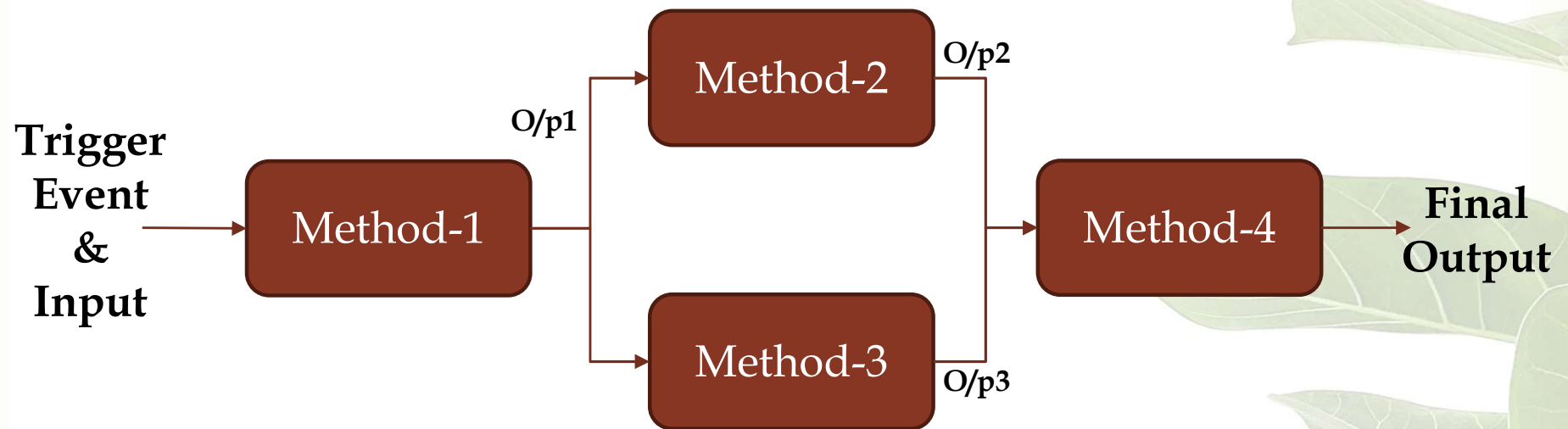
Previously
Proven
Method

A Methodology

Research Methodology



Time

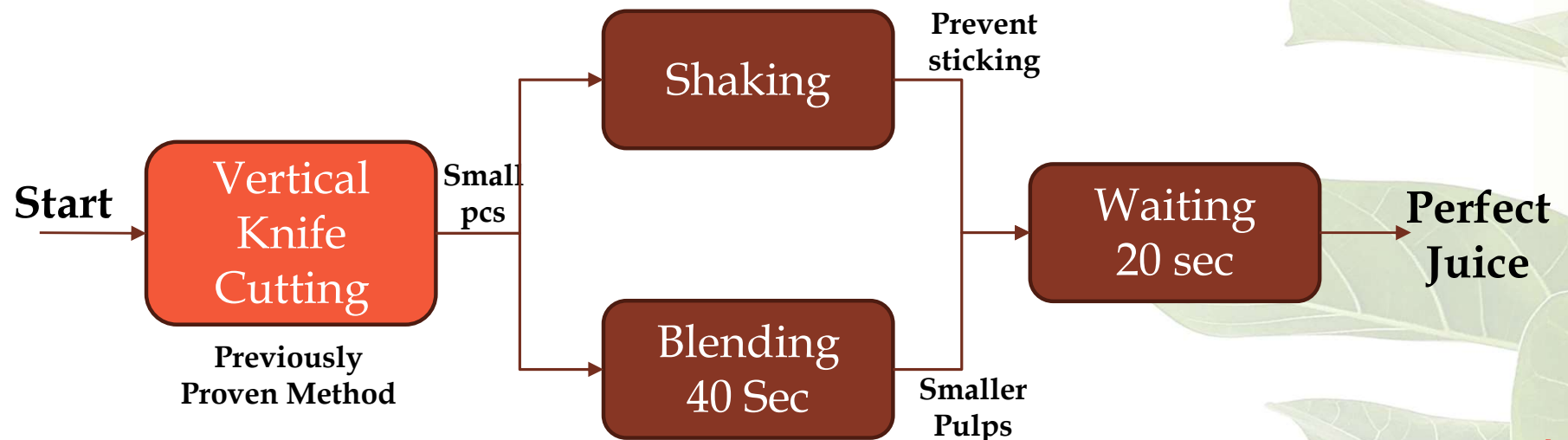


A Methodology

Example



Time →



A Methodology

Now...



- ❧ How good our newly developed methodology to achieve the research aim compares to the previously proven methods?
- ❧ How much time is needed by our new methodology to get the perfect juice?
- ❧ Is the cost of our new methodology lower than the previously proven methods?
- ❧etc. Many other questions, those must be answered by the next section....**"Research Scope"**



Research Scope

Research Scope



1. What does the research cover? And what it does NOT cover?
2. Limitations and Delimitations
3. Up-sides and Down-sides.
4. Until what extent, the proposed research work would solve the defined problem(s)?

This research would consider solving the problem partially
as the two guys will not be fully quenched.

The proposed methodology can be used for dividing ONLY
ONE apple at a time.

The proposed methodology works with the assumption that
the apple seeds were taken off.

Research Questions



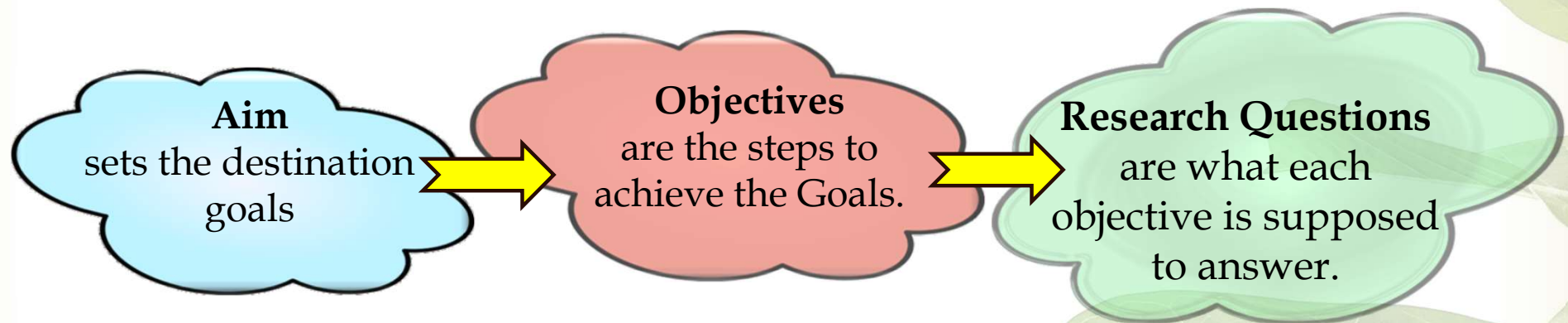
Will The Chosen Methodology help in achieving the research aims?

How to integrate a Chosen Methodology to the exist on?

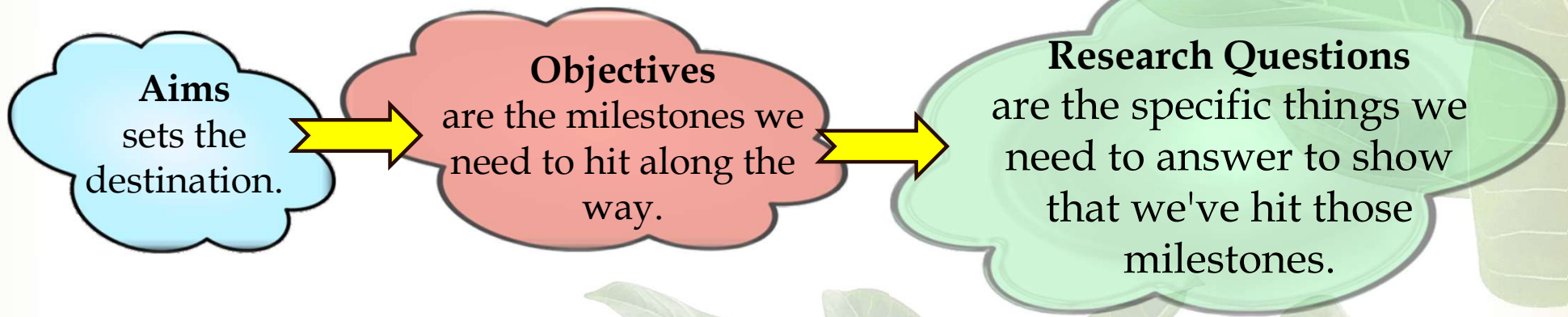
So How can we measure whether the answers for the research questions were positive or negative?

Every topic has **Performance Measures**, Basically, The Research Questions would ask whether the Chosen Methodology affect or influence those Performance Measures positively or not?

Research Questions



OR



The Relation between Research Questions and Research Objectives



Example Verb	Research Objective	Research Question
Evaluate	To evaluate the effectiveness of X.	➡ How effective is X?
Identify	To identify the factors affecting X.	➡ What factors affect X?
Compare	To compare X and Y.	➡ How does X compare with Y?
Develop	To develop a model for predicting X.	➡ How can a model be developed to predict X?
Integrate	To integrate X into Y.	➡ How can X be integrated into Y?

The Relation between Research Questions and Research Objectives



Example Verb	Research Objective	Research Question
Investigate	RO1: To investigate the relationship between digital literacy and academic performance.	RQ1: What is the relationship between digital literacy and academic performance?
Design	RO2: To design a framework for automated assessment of students' cognitive levels.	RQ2: How can a framework be designed to support automated assessment of students' cognitive levels?
Validate	RO3: To validate the proposed framework using expert evaluation.	RQ3: To what extent is the proposed framework valid according to expert evaluation?

Attached Full List Document: The Relation between Research Questions and Research Objectives

The Relation between Research Questions and Research Objectives



❧ IMPORTANT NOTE:

“The verb does not mechanically determine the question wording. What matters is that the meaning of the RQ matches the meaning of the RO.”

For example:

RO: To evaluate the effectiveness of the proposed system.

RQ: How effective is the proposed system?

The Relation between Research Questions and Research Objectives



Simple Golden rules:

- Every objective must be addressed by at least one question.
- But, a single objective can have two questions if it's multidimensional.
- Sometimes, one well-crafted question can address two very tightly linked objectives.
- However, for a master's or PhD thesis, one-to-one is the clearest and most common structure.

Bottom line is, don't chase numbers, just make sure that every objective is answerable via at least one question.

Part-1 Conclusion



Problem Statement



Research Aim(s)



Research Objectives



Methodology



Research Questions

Will The Chosen
Methodology help in
achieving the research
aims?

Research Scope

Part-1 Conclusion



We have a PROBLEM, that we AIM to solve through a set of OBJECTIVES those can be done as mentioned in details in the METHODOLOGY section.

The suggested METHODOGOLOGY has Limitations and Delimitations (SCOPE).

Now the QUESTION is; Will The Chosen Methodology help in achieving the research aims?



Exercise Time

Part-1 Exercise: One Page Research

You are required to come up with a daily life problem and analyze it in an academic way with its solution.

1. Problem Scenario Setup
2. Problem Statement
3. Aim(s)
4. Objective(s)
5. Methodology
6. Scope
7. Question(s)



End of PART-1